Nikhil Arora

Software Engineer | AI/ML Systems | Python Backend & Product Engineering
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SUMMARY

Recent NYU M.S. Computer Engineering graduate building Python backends and ML systems that remain measurable when data, workers, or assumptions fail. Developed a cross-platform distributed training workflow with worker recovery, a reviewable LLM/RAG interface over 10,000 network flows, and financial-data workflows that reduced recurring processing effort by **30%**. Comfortable taking ambiguous technical problems through implementation, testing, measurement, debugging, and clear handoff.

EDUCATION

New York University **May 2026**
M.S. Computer Engineering, GPA: 3.53 / 4.00 *New York, NY*
Coursework: Data Structures & Algorithms, Big Data Systems, Machine Learning, Deep Learning

CVM University – MBIT **May 2024**
B.E. Computer Engineering, GPA: 3.64 / 4.00 *India*

TECHNICAL SKILLS

Languages: Python, SQL, Java, Bash; JavaScript/TypeScript fundamentals
Backend & Product: FastAPI, Flask, REST APIs, Streamlit, PostgreSQL, SQLAlchemy, JSON/CSV services, React fundamentals
ML & Systems: PyTorch, PyTorch Distributed, CUDA, `torch.compile`, Hugging Face, scikit-learn, RAG, embeddings, Chroma
Infrastructure & Quality: Docker, Linux, Git/GitHub, pytest, Gloo, Tailscale, Spark/PySpark, logging, data validation

PROJECTS

Heterogeneous Distributed ML Training *Python, PyTorch Distributed, Gloo, Tailscale | GitHub*
- Built a leader-worker workflow across macOS and Windows with registration, heartbeats, rank assignment, reconnection, and synchronized training.
- Used per-worker benchmarks for Tiny ImageNet-200 shard allocation and evaluated ResNet-101 under Top-K gradient compression and synthetic straggler delays.

Net-RAG: Reviewable Open-Set Intrusion Detection *Python, scikit-learn, Chroma, Ollama, Streamlit | GitHub*
- Combined calibrated classifiers, retrieval over **10,000 network flows**, local Llama 3.1 explanations, deterministic guardrails, and a Streamlit review interface.
- Held one attack family out of training, added a shuffled-label leakage check, and raised its recall from **1.6% to 68.8%** at the selected threshold.

GPU-Efficient BERT Fine-Tuning *PyTorch, CUDA, Hugging Face, `torch.compile` | GitHub*
- Benchmarked BERT-base on SQuAD v2 using a 16 GB NVIDIA T4 across SDPA, compilation, padding, parameter-efficient tuning, and memory-saving methods.
- Increased throughput from **43.9 to 51.6 samples/s**, improved dev F1 from **76.6 to 77.2**, and reduced peak allocated VRAM by up to **30%**.

EXPERIENCE

NYU Stern, Endless Frontier Labs **Jul 2025 – Sep 2025**
Digital Tech Startup Analyst *New York, NY*
- Evaluated **50+ AI and software startups**, conducting deeper diligence on selected companies across architecture, data infrastructure, product capability, scalability, and execution risk.
- Translated technical and quantitative findings into concise recommendations for program stakeholders, separating demonstrated capabilities from assumptions and unresolved risks.

TD Tech Solutions **Jan 2024 – Apr 2024**
Software Developer Intern, Data Systems *India*
- Built a Python and SQL workflow for financial records covering ingestion, normalization, validation, and standardized JSON/CSV outputs.
- Refactored parsing and validation into reusable Flask modules, reducing recurring processing effort by **30%**.
- Added checks for missing, duplicate, malformed, and inconsistent records and reconciliation failures; documented interfaces and recovery procedures.

Tech Elecon **May 2023 – Aug 2023**
IoT & Data Engineering Intern *India*
- Cleaned and aligned high-frequency equipment telemetry and built diagnostic views for missing readings, throughput regressions, transmission failures, and abnormal operating patterns.